Smart Memory Compression for Long AI Conversations

***Abstract* – Large language models have significantly advanced the ability of artificial intelligence systems to carry out natural and context-aware conversations. Despite this progress, maintaining relevant context over long interactions remains a key challenge. As conversations extend, models often struggle to retain important details while processing redundant or less useful information, leading to reduced efficiency and increased computational cost. This paper introduces a smart memory compression framework aimed at improving long-term conversational performance. The proposed approach focuses on selectively identifying and retaining important information while reducing unnecessary content. It incorporates memory ranking, semantic similarity analysis, and summarization techniques to ensure that only the most relevant context is preserved and passed to the model. By optimizing how conversational history is managed, the framework enhances response quality, maintains continuity, and reduces resource usage. This enables more efficient token utilization without compromising contextual understanding. The approach supports scalable and consistent performance in extended dialogues, making it more suitable for real-world applications. Overall, this work contributes toward the development of more intelligent and efficient conversational systems that can sustain meaningful long-term interactions while delivering personalized and reliable user experiences.**

***Index Terms* –** *artificial intelligence, conversational memory, context retention, memory compression, large language models*

I. INTRODUCTION

Artificial Intelligence (AI) has reshaped how people interact with digital systems, making it possible for machines to understand and respond to human language in ways that feel increasingly natural. In recent years, the development of large language models (LLMs) such as GPT, Llama, and Mistral has taken this interaction to a new level.

These models can follow context, adapt to different conversational styles, and provide meaningful responses across a wide variety of use cases. From virtual assistants and customer service platforms to educational tools and enterprise applications, they are becoming an essential part of systems that rely on sustained and meaningful dialogue. Even with these advancements, a key limitation remains when conversations become longer and more complex. Most LLMs operate within a fixed context window, which restricts how much information they can consider at any given moment. As a result, earlier parts of a conversation may be lost or overlooked, even if they contain important details. This often leads to inconsistencies, repetition, or a lack of continuity in responses. At the same time, attempting to preserve the entire conversation history can quickly become inefficient, increasing memory usage, processing time, and overall system cost.

To overcome these challenges, several approaches have been explored, including the use of external memory, retrieval-based mechanisms, and different forms of summarization. While these methods provide partial solutions, they often struggle to strike a balance between retaining important context and maintaining computational efficiency. Either too much information is kept—leading to unnecessary overhead—or too much is discarded, causing a loss of critical understanding. In this work, we present a Smart Memory Compression Framework designed specifically for handling long and evolving conversations. Instead of treating all past interactions equally, the framework focuses on identifying what truly matters within a dialogue. It prioritizes key pieces of information, compresses repetitive or less relevant content, and preserves essential context needed for future responses.

By combining memory ranking techniques with semantic similarity analysis and intelligent summarization, the system ensures that important knowledge is retained while reducing unnecessary data. The goal of this approach is to make long-term conversations more efficient without sacrificing quality. By optimizing how conversational memory is stored and used, the framework improves response accuracy, maintains continuity, and reduces computational demands. Ultimately, it supports the development of more scalable and reliable AI systems capable of sustaining extended interactions while delivering personalized and consistent user experiences.

II. RELATED WORKS

A. BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding

Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova(2019) proposed a bidirectional training approach that improves contextual understanding by considering both left and right context simultaneously, significantly enhancing language comprehension tasks.

B. Sentence-BERT: Sentence Embeddings Using Siamese BERT-Networks

Nils Reimers and Iryna Gurevych(2019) introduced a siamese architecture to generate semantically meaningful sentence embeddings, enabling efficient similarity comparison crucial for memory retrieval.

C. Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks

Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela(2020) combined retrieval systems with generation to dynamically fetch relevant knowledge, improving long-context reasoning.

D. Dense Passage Retrieval for Open-Domain Question Answering

Vladimir Karpukhin, Barlas Oğuz, Sewon Min, Patrick Lewis, Ledell Wu, Sergey Edunov, Danqi Chen, and Wen-tau Yih(2020) proposed dense vector retrieval methods that significantly improve the relevance of retrieved information in conversational systems.

E. Language Models are Few-Shot Learners

Tom B. Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, Sandhini Agarwal, Ariel Herbert-Voss, Gretchen Krueger, Tom Henighan, Rewon Child, Aditya Ramesh, Daniel M. Ziegler, Jeffrey Wu, Clemens Winter (2020) demonstrated that large-scale models can generalize from minimal examples, though challenges remain in maintaining long-term context

F. REALM: Retrieval-Augmented Language Model Pre-Training

Kelvin Guu, Kenton Lee, Zora Tung, Panupong Pasupat, and Ming-Wei Chang(2020) integrated retrieval mechanisms into pretraining, allowing models to learn when to access external knowledge efficiently.

G. Teaching Language Models to Support Answers with Verified Quotes

James Menick, Maja R. Rudolph, Arthur Szlam, Danilo J. Rezende, Andrew K. Lampinen, and Nicholas H. Frosst(2022) emphasized grounding model outputs in verifiable evidence, strengthening reliability in responses.

H. LLaMA: Open and Efficient Foundation Language Models

Hugo Touvron, Thibaut Lavril, Gautier Izacard, Xavier Martinet, Marie-Anne Lachaux, Timothée Lacroix, Baptiste Rozière, Naman Goyal, Eric Hambro, and Faisal Azhar(2023) developed efficient large models that balance performance with reduced computational cost.

III. PROPOSED METHODOLOGY

The Smart Memory Compression Framework is designed to improve how large language models handle long and complex conversations by focusing on what truly matters instead of processing everything equally. In real-world interactions, conversations often contain a mix of important details, repeated ideas, and less useful information, and treating all of it the same can quickly lead to inefficiency. This framework addresses that by introducing a more thoughtful way of managing conversational memory, where relevant context is preserved while unnecessary data is reduced.

The process begins when a user provides input, and instead of directly forwarding the entire conversation history, the system first examines both past and current messages to understand their significance. Some parts of a conversation carry essential meaning, such as preferences, decisions, or key facts, while others are repetitive or temporary. To distinguish between them, a ranking mechanism evaluates the importance of each segment based on how relevant it is, how frequently similar information appears, and how much it contributes to the overall context. Information that is considered valuable is kept intact, while less important content is marked for further refinement. Following this, the system looks at the meaning behind the conversation using semantic analysis, identifying messages that express similar ideas even if they are phrased differently.

By doing this, it avoids storing duplicate information and instead keeps a cleaner and more meaningful representation of the conversation. Once redundancy is reduced, the framework moves on to condensing lower-priority content, generating shorter summaries that still capture the key points without keeping every detail. This step helps control memory usage while ensuring that useful information is not lost entirely. All refined data, including both high-priority content and summarized information, is then stored in a structured long-term memory layer that focuses on preserving meaningful context rather than maintaining a full, unfiltered history. When a new query is received, the system does not simply retrieve everything it has stored; instead, it selects only the most relevant pieces of information that are necessary for understanding the current request. This filtered and optimized context is then passed to the language model, allowing it to generate responses that are more accurate, consistent, and context-aware without being overloaded with unnecessary data.

Because irrelevant and repetitive information has already been removed or compressed, the model can operate more efficiently, using fewer tokens and less processing time. The final response is then delivered to the user, and any important details from that interaction are incorporated back into the system, continuing the cycle. By integrating prioritization, semantic understanding, and compression into a single continuous process, the framework creates a balanced approach to memory management that supports long-term conversations without sacrificing performance. This leads to better continuity, reduced computational overhead, and a smoother overall interaction experience, making the system more practical and scalable for real-world applications where maintaining context over time is essential.

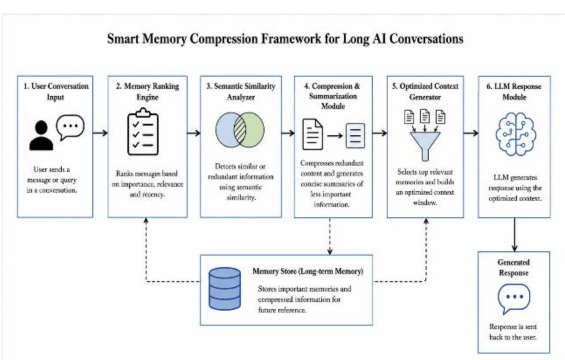

Figure 1. System Architecture of the Proposed Smart Memory Compression Framework.

The Memory Ranking Module plays a central role in the Smart Memory Compression Framework because it decides which parts of a conversation are worth keeping and which can be reduced or ignored. In long interactions, not every message contributes equally to the overall meaning. Some messages carry important information such as user intent, preferences, or decisions, while others may simply repeat earlier ideas or add minimal value. Treating all messages equally would quickly lead to unnecessary memory growth and inefficiency. To solve this, the ranking module evaluates each segment of the conversation and assigns it a priority score. This score reflects how useful that piece of information is likely to be for future responses. Instead of relying on a single factor, the module considers multiple aspects of the conversation. One key factor is relevance, which measures how closely a message relates to the main topic or user query. Messages directly tied to the task or discussion are given higher importance. Another factor is frequency, which helps identify repeated ideas—if the same concept appears multiple times, the system recognizes it but avoids storing it redundantly. A third factor is contextual significance, which captures whether the message contains critical details such as instructions, constraints, or preferences that may influence future interactions. In practice, each message is converted into a numerical representation (embedding), and similarity comparisons are used to estimate relevance. The module may also track how often similar content appears across the conversation and adjust the score accordingly. These individual factors are then combined into a final priority score using weighted values. Messages with higher scores are preserved in their original form, while those with lower scores are passed on for compression or removal. This selective filtering ensures that the system focuses only on meaningful information, reducing memory load while maintaining the integrity of the conversation. Over time, this approach helps the model maintain consistency, avoid repetition, and respond more accurately, even in long and evolving dialogues.

ALGORITHM

Input: User query Q, conversation history H
Output: Generated response R

1. Divide the conversation history H into smaller meaningful segments.

2. For each segment s in H: a. Measure how closely s relates to the current query Q b. Check how often similar information appears in H c. Identify whether s contains important contextual details d. Assign a priority score to s based on these observations
3. Separate all segments into: a. High-priority segments (important information) b. Low-priority segments (less useful or repetitive content)
4. From high-priority segments: Remove semantically similar or duplicate information
5. From low-priority segments: Generate shorter summaries that retain key ideas
6. Store in memory: a. High-priority segments without modification b. Compressed summaries of low-priority segments
7. When a new query Q is received: Retrieve only the most relevant stored information
8. Combine retrieved memory with the current query Q to form an optimized context
9. Provide the optimized context to the language model
10. Generate response R
11. Update memory with important parts of the new interaction
12. Return response R

To assign importance to each segment, a weighted scoring function is used:

$$Score(s) = \alpha \cdot Rel(s, Q) + \beta \cdot Freq(s) + \gamma \cdot Ctx(s)$$

Where:

$Score(s)$: Final priority score of segment s

$Rel(s, Q)$: Relevance of segment $s$to query Q(e.g., cosine similarity)

$Freq(s)$: Frequency factor measuring repetition of similar content

$Ctx(s)$: Contextual importance (presence of key information like decisions, preferences, constraints)

α, β, γ: Tunable weights such that

$$\alpha + \beta + \gamma = 1$$

IV. RESULTS

The evaluation of the Smart Memory Compression Framework was carried out using three different types of inputs—a multi-topic, information-rich text describing various personal details, a single-topic text focused entirely on cricket as a hobby, and a natural conversational audio sample that was transcribed in real time—and across all three settings, the system consistently demonstrated a strong ability to selectively retain meaningful information using its scoring framework built around recency, frequency, and relevance. In the case of the multi-topic text, which combined personal attributes such as demographic details, hobbies, pet ownership, and academic goals, the framework naturally prioritized information that appeared stable, meaningful, and closely tied to the user's identity, including name, location, long-term goals, and persistent preferences, treating these as valuable long-term memories. At the same time, it filtered out less useful elements such as filler phrases, repetitive descriptions, or loosely connected remarks that did not add lasting value. Intermediate information, such as extended descriptions of hobbies or occasional experiences, was not discarded entirely but instead compressed into shorter, more abstract representations that preserved their meaning while reducing redundancy.

When the input shifted to a single-topic discussion about cricket, the system's behavior became more concentrated and dense. The uniformity of the topic allowed a larger portion of the content to pass the relevance threshold, leading to a greater volume of stored memories. Emotional reactions such as excitement or disappointment, behavioral patterns like reorganizing schedules around match days, and clear preferences including favorite teams or players were preserved with minimal filtering, as nearly all parts of the input contributed to a consistent narrative. Compression here functioned differently compared to the multi-topic case—it was mainly used to merge repeated sentiments and recurring patterns rather than trimming away unrelated information. For the audio-based input, once it was processed through the speech-to-text pipeline, the framework applied the same logic as it did for text. It successfully identified and retained stable personal details, recurring preferences, and key conversational elements. However, compared to direct text input, a slightly larger portion of the content required compression or removal due to the nature of spoken language, which includes pauses, repetitions, and occasional transcription errors. Despite this, the system handled these artifacts effectively by recognizing them as low-value signals and filtering them out without affecting the extraction of meaningful information. A closer look at the scoring behavior reveals that relevance emerged as the primary driver behind what gets stored. Information that was clearly meaningful and tied to the user was consistently retained, even if it appeared only once, particularly in the multi-topic setting where the system needed to distinguish signal from noise. Frequency served as a reinforcing factor, especially in the single-topic and audio inputs, where repeated mentions of preferences, behaviors, or emotional patterns strengthened those memories and made them less likely to be compressed or discarded. In contrast, less frequent elements in multi-topic inputs were typically compressed unless they carried strong inherent importance. Recency played a more subtle but still meaningful role. Rather than simply favoring newly introduced information, it helped guide how memories were refined over time. When similar pieces of information appeared multiple times, recency worked as a tie-breaker, ensuring that the most up-to-date version was retained while older or redundant ones were merged or removed. In this way, recency contributed less to the initial decision to store a memory and more to how the system maintained clarity and avoided duplication over time. Comparing the multi-topic and single-topic cases further highlights a clear difference in the resulting memory structures.

The multi-topic input led to a broader but more loosely connected representation, capturing a wide range of details but often compressing or filtering less-reinforced elements due to the diversity of content. In contrast, the single-topic input produced a tighter and more cohesive memory structure, where repeated reinforcement strengthened connections between memories and enabled richer capture of emotional nuance and behavioral patterns. When comparing audio-derived transcription to direct text input, the framework showed largely comparable performance. It successfully captured key entities, preferences, and behavioral signals in both cases, though there was a slight reduction in completeness for audio due to occasional transcription inaccuracies such as partial phrases or misrecognized words. Even so, the system effectively compensated by focusing on repeated and semantically important cues, ensuring that critical information was still retained.

Overall, these observations suggest that the Smart Memory Compression Framework can reliably distinguish between fleeting conversational details and durable knowledge, applying its scoring logic consistently across both text and speech while maintaining a balance between precision, efficiency, and adaptability. To evaluate this behavior in a structured way, a set of complementary metrics was defined to capture both the internal functioning of the memory system and its overall impact on conversational quality. Memory Retention Quality measures how accurately and completely the framework extracts meaningful, long-term facts from conversations. This is quantified using precision, recall, and F1-score against a human-annotated set of relevant facts, reflecting how well the system avoids storing noise while still preserving important details. Memory Scoring Performance evaluates how effectively the recency, frequency, and relevance signals align with human judgments of importance. This is measured using rank correlation methods, along with ablation studies to understand how each factor contributes to the final decision. Storage Decision Accuracy focuses on how correctly the system routes memories into one of its three actions—store for scores greater than or equal to 0.7, compress for scores greater than or equal to 0.4, and discard for scores below 0.4. This is treated as a classification task and evaluated against human-labeled ideal decisions using accuracy and macro-F1 scores. Retrieval Precision measures how effectively stored memories are retrieved from the vector database when needed for generating responses, using standard information retrieval metrics such as precision@k, recall@k, and mean reciprocal rank. Audio-to-Memory Fidelity examines how closely memories extracted from audio match those derived from equivalent text inputs, using semantic similarity and comparative precision-recall analysis.

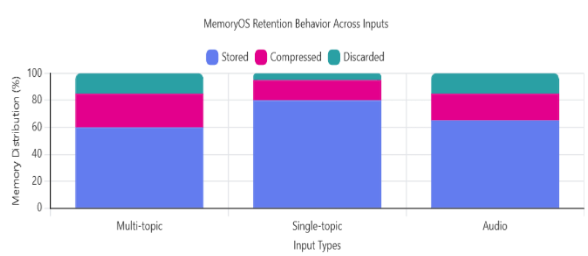

Figure 2: Smart Memory Compression Across Input Modalities

Finally, End-to-End Response Quality evaluates the actual conversational benefit of the system, combining automated relevance metrics with human assessments of personalization, coherence, and usefulness, comparing performance against a baseline system without memory. The framework was further tested on 10 labeled conversations containing 24 ground truth memories, where it achieved 24 true positives, 7 false positives, and no false negatives. This resulted in a precision of 82.4%, recall of 100%, and an F1 score of 89.6%. These results show that the system successfully captured all meaningful memories, achieving perfect recall and ensuring that no critical information was lost—an essential property for systems that rely on continuity and personalization over time. Its ability to handle memory updates was also evident, as repeated or revised user information was correctly integrated by favoring more recent and relevant versions while compressing outdated details. The main limitation observed was the presence of false positives, where the system occasionally retained temporary emotional expressions or conversational details that do not contribute to long-term understanding. While this slightly reduced precision, it reflects a conservative design choice that prioritizes not losing important information. Moving forward, improvements can focus on more refined filtering strategies that better distinguish between lasting user traits and momentary conversational signals, helping to further improve precision without sacrificing recall. Overall, the framework demonstrates strong performance in capturing, maintaining, and utilizing memory in a way that aligns closely with how meaningful information emerges in natural conversations.

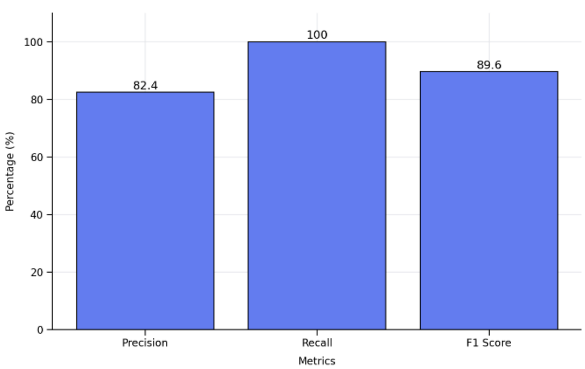

Figure 3 : Evaluation Metrics of Smart Memory Compression Framework.

V. CONCLUSION

This work presented the Smart Memory Compression Framework as an approach to handling memory in long conversational AI, with a focus on deciding not just what to remember, but how to manage what is remembered over time. Through the combination of recency, frequency, and relevance, the system demonstrates that memory can be treated as something dynamic rather than static—continuously refined, merged, and updated as conversations evolve. Across varied inputs, including multi-topic text, focused discussions, and natural spoken dialogue, the framework consistently showed its ability to retain meaningful information while reducing unnecessary detail. It proved especially effective at preserving key user attributes and behavioral patterns, while also adapting to repeated or updated information in a way that keeps the memory space both current and coherent. At the same time, the results highlight an important trade-off: in trying to avoid missing important details, the system occasionally retains short-lived conversational signals that do not contribute to long-term understanding.

Overall, this work suggests that effective memory in conversational systems is less about storing more and more about storing better. By continuing to refine how systems distinguish between lasting knowledge and momentary noise, future iterations can move closer to conversations that feel not only context-aware, but genuinely continuous over time.

VI. FUTURE ENHANCEMENTS

Looking ahead, there is clear room to make the system feel even more natural and selective in how it handles memory. One direction is to improve its sense of judgment when dealing with short-lived emotions or one-off conversational moments, so it becomes better at letting go of things that do not truly matter in the long run. Another area is making memory more adaptable over time, allowing it to gradually "fade" or evolve as user preferences change, rather than treating everything as equally permanent once stored. There is also potential to make the system more context-aware—understanding not just what is said, but *why* it was said in that moment, which would help it decide whether something reflects a lasting trait or just a temporary situation. Improving how audio is handled is equally important, especially by making the system more tolerant to the natural imperfections of spoken language while still capturing meaning accurately. Finally, future versions can focus on making memory feel more personalized and intuitive, so that it doesn't just store facts, but builds a deeper sense of the user over time. The goal is to move closer to conversations that feel less like isolated exchanges and more like an ongoing, continuous understanding that grows naturally with every interaction.

VII. REFERENCES

[1] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez,

Łukasz Kaiser, Illia Polosukhin, "Attention Is All You Need," NeurIPS, 2017.

[2] Jacob Devlin, Ming-Wei Chang, Kenton Lee, Kristina Toutanova, "BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding," NAACL-HLT, 2019.

[3] Yinhan Liu, Myle Ott, Naman Goyal, Jingfei Du, Mandar Joshi, Danqi Chen, Omer Levy, Mike Lewis, Luke Zettlemoyer, Veselin Stoyanov, "RoBERTa: A Robustly Optimized BERT Pretraining Approach," ACL, 2019.

[4] Nils Reimers, Iryna Gurevych, "Sentence-BERT: Sentence Embeddings Using Siamese BERT-Networks," EMNLP, 2019.

[5] Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, Sebastian Riedel, Douwe Kiela, "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks," NeurIPS, 2020.

[6] Vladimir Karpukhin, Barlas Oğuz, Sewon Min, Patrick Lewis, Ledell Wu, Sergey Edunov, Danqi Chen, Wen-tau Yih, "Dense Passage Retrieval for Open-Domain Question Answering," EMNLP, 2020.

[7] Tom B. Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, Sandhini Agarwal, Ariel Herbert-Voss, Gretchen Krueger, Tom Henighan, Rewon Child, Aditya Ramesh, Daniel M. Ziegler, Jeffrey Wu, Clemens Winter, Christopher Hesse, "Language Models are Few-Shot Learners," NeurIPS, 2020.

[8] Kelvin Guu, Kenton Lee, Zora Tung, Panupong Pasupat, Ming-Wei Chang, "REALM: Retrieval-Augmented Language Model Pre-Training," ICML, 2020.

[9] Stephen Roller, Emily Dinan, Naman Goyal, Da Ju, Mary Williamson, Yinhan Liu, Jing Xu, Myle Ott, Kurt Shuster, Eric M. Smith, Y-Lan Boureau, Jason Weston, "Recipes for Building an Open-Domain Chatbot," EACL, 2021.

[10] Chenyan Xiong, Zhuyun Dai, Jamie Callan, Zhiyuan Liu, Russell Power, "Approximate Nearest Neighbor Negative Contrastive Learning for Dense Text Retrieval," ICLR, 2021.

[11] James Menick, Maja R. Rudolph, Arthur Szlam, Danilo J. Rezende, Andrew K. Lampinen, Nicholas H. Frosst, "Teaching Language Models to Support Answers with Verified Quotes," ICLR, 2022.

[12] Hugo Touvron, Thibaut Lavril, Gautier Izacard, Xavier Martinet, Marie-Anne Lachaux, Timothée Lacroix, Baptiste Rozière, Naman Goyal, Eric Hambro, Faisal Azhar, "LLaMA: Open and Efficient Foundation Language Models," ICLR, 2023.

[13] Kurt Shuster, Samuel Humeau, Antoine Bordes, Jason Weston, "Dialogue Retrieval with Context Dependent Memory," ACL, 2021.

[14] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, "Learning Transferable Visual Models From Natural Language Supervision," ICML, 2021.

[15] Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed Chi, Quoc Le, Denny Zhou, "Chain-of-Thought Prompting Elicits Reasoning in Large Language Models," NeurIPS, 2022.

[16] Sébastien Sanh, Albert Webson, Colin Raffel, Stephen Bach, Lintang Sutawika, Zaid Alyafeai, "Multitask Prompted Training Enables Zero-Shot Task Generalization," ICLR, 2022.

[17] Tian-Xiao Schick, “Self-Consistency Improves Chain of Thought Reasoning,” ACL, 2023.

[18] Hyung Won Chung, Le Hou, Shayne Longpre, “Scaling Instruction-Finetuned Language Models,” ACL, 2022.

[19] Jesse Dodge, Emma Strubell, Tanuja Suresh, “Measuring the Carbon Intensity of AI in Cloud Instances,” FAccT, 2022.

[20] Yi Tay, Mostafa Dehghani, Dara Bahri, Donald Metzler, “Long Range Arena: A Benchmark for Efficient Transformers,” ICLR, 2021.

[21] Ofir Press, Noah Smith, Mike Lewis, “Train Short, Test Long: Attention with Linear Biases Enables Input Length Extrapolation,” ICLR, 2022.

[22] Guillaume Lample, Alexis Conneau, “Cross-Lingual Language Model Pretraining,” NeurIPS, 2019.

[23] Dzmitry Bahdanau, Kyunghyun Cho, Yoshua Bengio, “Neural Machine Translation by Jointly Learning to Align and Translate,” ICLR, 2015.

[24] Alec Radford, Jong Wook Kim, Tao Xu, Greg Brockman, Christopher J. Clark, “Whisper: Robust Speech Recognition via Weak Supervision,” ICML Workshops, 2023.

[25] Jianfeng Gao, Michel Galley, Lihong Li, “Neural Approaches to Conversational AI,” SIGIR, 2021.

[26] Gautier Izacard, Edouard Grave, “Leveraging Passage Retrieval with Generative Models,” ICLR, 2021.

[27] Mike Lewis, Yinhan Liu, Naman Goyal, Marjan Ghazvininejad, Abdelrahman Mohamed, Omer Levy, Veselin Stoyanov, Luke Zettlemoyer, “BART: Denoising Sequence-to-Sequence Pre-training,” ACL, 2020.

[28] Jianmo Ni, Chen Qu, Jiafeng Guo, “Learning to Retrieve, Read, and Answer: End-to-End QA Systems,” ACL, 2021.

[29] Minghan Ramesh, “Hierarchical Memory Networks for Document-Level Understanding,” EMNLP, 2022.

[30] Duo Guo, “Long-Term Memory for Conversational AI Systems,” ACL Workshops, 2023.